PINNStudio, tuned (forward, inverse) | Starting settings: DeepXDE's example / the paper's method | Both the same (no tuning needed) | Reported in the paper

ODEs
- Integro-differential ODE
- Integro-differential ODE – Volterra kernel
- Linear ODE system — ω
- Lotka-Volterra ODE — a, d
- Second-order ODE — a, b
- Lorenz ODE – unknown C_1, C_2, C_3 — C_3, C_2, C_1
- Lorenz ODE – unknown C_1, C_2, C_3, exogenous input — C_3, C_2, C_1

1D, steady
- Euler beam 1D — EI
- Fractional Poisson 1D
- Poisson 1D – Dirichlet
- Poisson 1D – Dirichlet/Neumann
- Poisson 1D – Dirichlet/Periodic
- Poisson 1D – Dirichlet/point-set operator
- Poisson 1D – Dirichlet/Robin
- Poisson 1D – hard BCs
- Poisson 1D – multi-scale Fourier features
- Brinkman-Forchheimer 1D – unknown v_e, K — K, ν_e
- Diffusion-reaction 1D – unknown rate field — $k(x)$
- Fractional Poisson 1D – unknown order α — α
- Poisson 1D – unknown forcing field — $q(x)$
- Poisson 1D – unknown forcing field (batches) — $q(x)$

1D, time-dependent
- Allen-Cahn 1D – Wight & Zhao — γ_2, γ_1
- Cahn-Hilliard 1D – Wight & Zhao
- Cahn-Hilliard 1D – Mattey & Ghosh
- Black-Scholes 1D – American put
- Black-Scholes 1D – European call
- Burgers 1D – Raissi et al. — λ_1, λ_2
- Burgers 1D – RAR
- Diffusion 1D
- Diffusion 1D – hard IC and BCs
- Diffusion 1D – resampling
- Diffusion-reaction 1D — D
- Fractional diffusion 1D
- Heat 1D – Dirichlet — a
- Heat 1D – resampling
- Klein-Gordon 1D — γ
- Schrödinger 1D — c
- Wave 1D – space-time multi-scale Fourier features — C
- Diffusion 1D – unknown coefficient — C
- Diffusion-reaction 1D – unknown k_f, D — D, kf

2D, steady
- Fractional Poisson 2D – disk
- Helmholtz 2D – Dirichlet — k_0
- Helmholtz 2D – hyperparameter search
- Helmholtz 2D – sound-hard scattering
- Helmholtz 2D – square with hole
- Laplace 2D – disk
- Linear elasticity 2D – plate — μ, λ
- Linear elasticity 2D – plate, soft BCs
- Navier-Stokes 2D – Kovasznay flow — Re
- Poisson 2D – disk — f
- Poisson 2D – L-shape
- Fractional Poisson 2D – unknown order α — α

2D, time-dependent
- Allen-Cahn 2D – Wight & Zhao — $\lambda, \lambda\varepsilon^2$
- Allen-Cahn 2D – Mattey & Ghosh — c_2, c_1^2
- Cahn-Hilliard 2D – Wight & Zhao — ε^2
- Cahn-Hilliard 2D – Mattey & Ghosh — kappa, α
- Burgers 2D – Mathias et al. — ν
- Heat 2D — α

3D
- Fractional Poisson 3D – sphere
- Poisson 3D – cube with spherical cut-out
- Poisson 3D – sphere — f
- Allen-Cahn 3D – Wight & Zhao — $\lambda\varepsilon^2, \lambda$
- Heat 3D — α
- Navier-Stokes 3D – Beltrami flow — ν

10^{-6} 10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 10^{0}

Relative L_2 error of the main output at the last reference time

0.001 0.1 10

Parameter (or unknown field) error (%)